2.2 Answers

I will be using an RNN model since the CNN is better for spatial visual data, but the RNN can handle temporal data, which is a significant feature of the climate data set, and is used for weather analysis.

Test #1

Epochs = 10

N_hidden = 4

Accuracy increases from 9% to 29%, loss increases from 9.1 to 9.6

Starting hyperparameters

```
epochs = 10
batch_size = 16
n_hidden = 4

timesteps = len(X_train[0])
input_dim = len(X_train[0][0])
n_classes = y_train.shape[1]

model = Sequential()
model.add(LSTM(n_hidden, input_shape=(timesteps, input_dim)))
model.add(Dropout(0.5))
model.add(Dense(n_classes, activation='sigmoid')) #Don't use relu here!
```

```
print(confusion_matrix(y_test, model.predict(X_test)))

180/180 ━━━━━━━━━━━━━━━━━━━━ 2s 9ms/step
Pred         BASEL  BELGRADE  MADRID
True
BASEL         1491         1    2190
BELGRADE       981         0     111
BUDAPEST       207         0       7
DEBILT          82         0       0
DUSSELDORF      28         0       1
HEATHROW        68         0      14
KASSEL          11         0       0
LJUBLJANA       52         0       9
MAASTRICHT       4         0       5
MADRID         177         0     281
MUNCHENB         7         0       1
OSLO             4         0       1
STOCKHOLM        4         0       0
VALENTIA         0         0       1
```

Starting confusion matrix – this represents the low accuracy with data points spread between Basel and Madrid, and it cannot recognise Sonnblick, likely due to the answer data only having 0s.

Test #2

Epochs = 20

N_hidden = 8

Accuracy increases from 5% to 52%, loss increases from 9.03 to 11.98. While the loss increases, the accuracy is much better compared to 4 layers.

Test #3 – to see how changing layers but keeping epochs makes a difference

Epochs = 20

N_hidden = 16

Accuracy increases from 11% to 64%, loss increases from 9.7 to 13.66. Overall better accuracy for the loss increasing.

Test #4 - to see how changing epochs but keeping layers makes a difference

Epochs = 30

N_hidden = 8

Accuracy increases from 11% to 59%, loss increases from 9 to 13.5

Changing hidden layers to 16 seems to have a better impact than epochs so will proceed with 20 epochs and increasing layers.

Test #5

Epochs = 20

N_hidden = 32

Accuracy begins at 8% and finishes at 65%, loss starts at 10 and ends at 16.3

This shows that increasing hidden layers from 16 to 32 has little effect on accuracy and increases the loss compared to test #3

Test #6 using convolution and pooling

Epochs = 20

N_hidden = 16

I added convolution and pooling and reduce the layers back to 16. Unfortunately, the accuracy stalled at 31% but finished on 65% on epoch 20, however the loss kept increasing from 9.7 to settle on 76!

As convolution and pooling and layers higher than 16 have had negative effects, I removed convolution and pooling, and based on the most successful 16 layers and 20 epochs, I tried out tanh to see if it would have a positive impact.

Test #7 – tanh activation

Epochs = 20

N_hidden = 16

Accuracy began at 4% but still only settled only on 10%, while the loss started at a much higher 24 and increased to 26 – not successful! As a final test, changed activation back to sigmoid, kept layers at 16 and increased epochs to 30.

Test #8

Epochs = 30

N_hidden = 16

Accuracy began at 8% but still only settled only on 65%, loss started at 9.7 and increased to 14.6.

Final hyperparameters

```
epochs = 30
batch_size = 16
n_hidden = 16

timesteps = len(X_train[0])
input_dim = len(X_train[0][0])
n_classes = y_train.shape[1]

model = Sequential()
model.add(LSTM(n_hidden, input_shape=(timesteps, input_dim)))
model.add(Dropout(0.5))
model.add(Dense(n_classes, activation='sigmoid')) #Don't use relu here!
```

Final confusion matrix

```
180/180 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step
```

Pred / True	BASEL	BELGRADE	MADRID
BASEL	3675	1	6
BELGRADE	1089	0	3
BUDAPEST	214	0	0
DEBILT	82	0	0
DUSSELDORF	29	0	0
HEATHROW	81	0	1
KASSEL	11	0	0
LJUBLJANA	61	0	0
MAASTRICHT	9	0	0
MADRID	449	1	8
MUNCHENB	8	0	0
OSLO	5	0	0
STOCKHOLM	4	0	0
VALENTIA	1	0	0

The model has 65% accuracy which is mostly made up of correctly guessing Basel, however it seems to have attributed nearly all the rest to Basel and just guessed 8 correctly for Madrid!

It has only recognised 3 stations so would likely need a different model to successfully predict for the other 12 stations, while appreciating Sonnblick will be limited by the nature of its data.